



Lightweight and Faithful Visual Condition Checking in Behavior Trees via Expert-Regularized Reinforcement Learning



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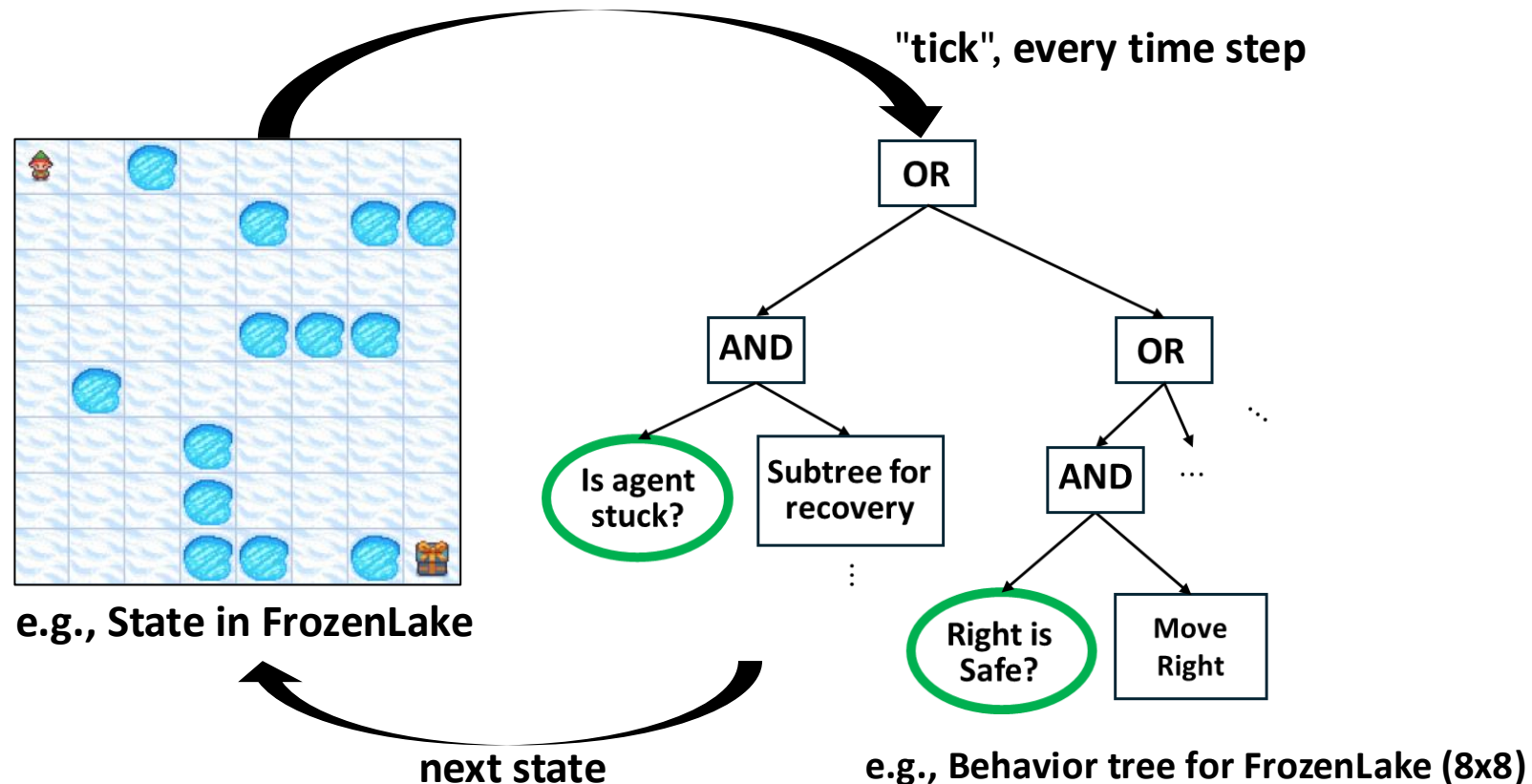
■ Behavior Trees

- **Behavior Tree¹** (BT) = A hierarchical control logic

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- **Behavior Tree¹** (BT) = A hierarchical control logic
- Human-readable, modular, reusable, and reactive



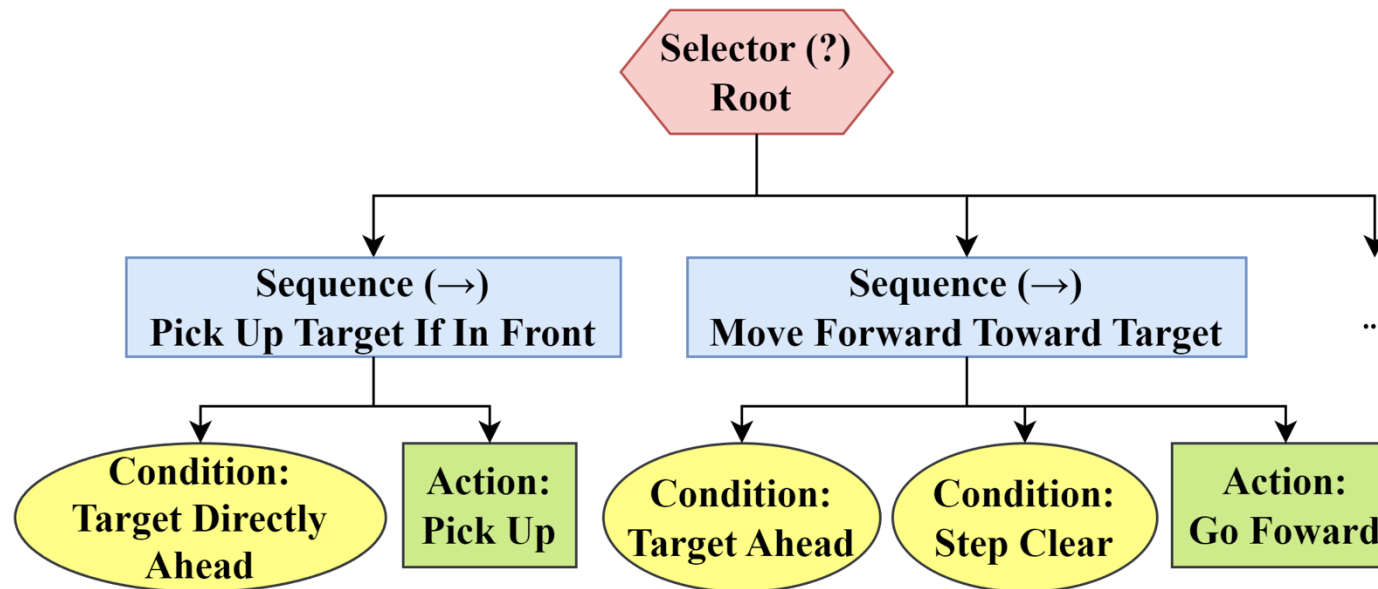
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- **Control Nodes** : Selector (?, OR), Sequence (→, AND),
- **Leaf Nodes** : Condition (ellipses), Action (rectangles),

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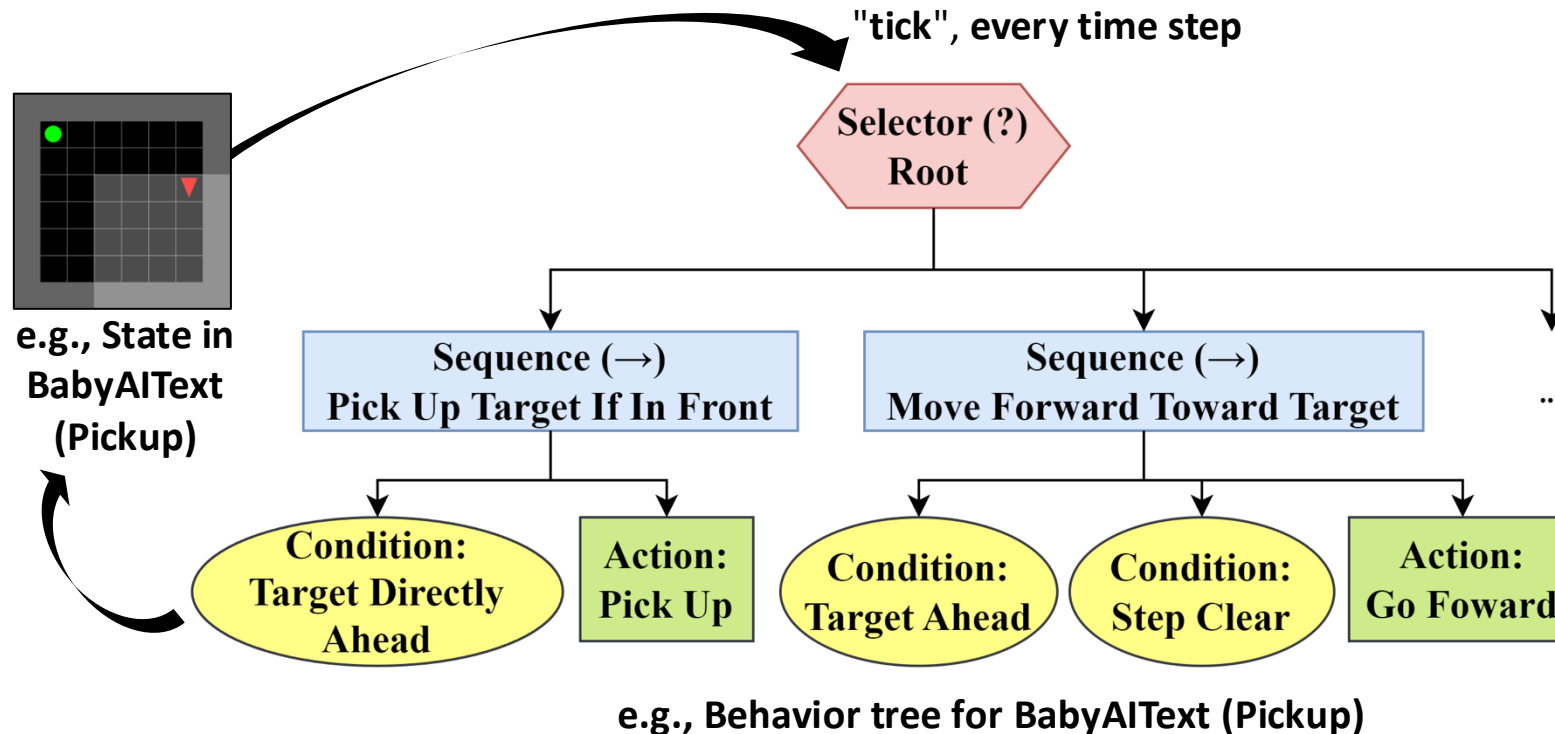
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e.g., Behavior tree for BabyAIText (Pickup)

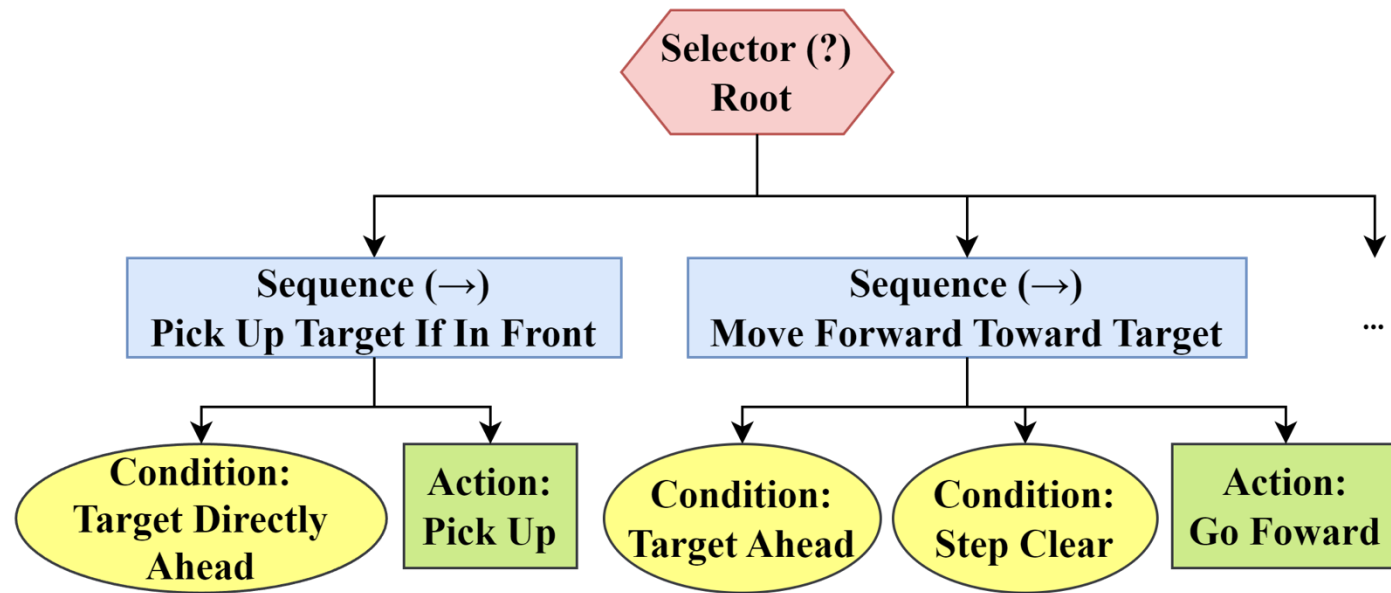
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■ Motivation for this research

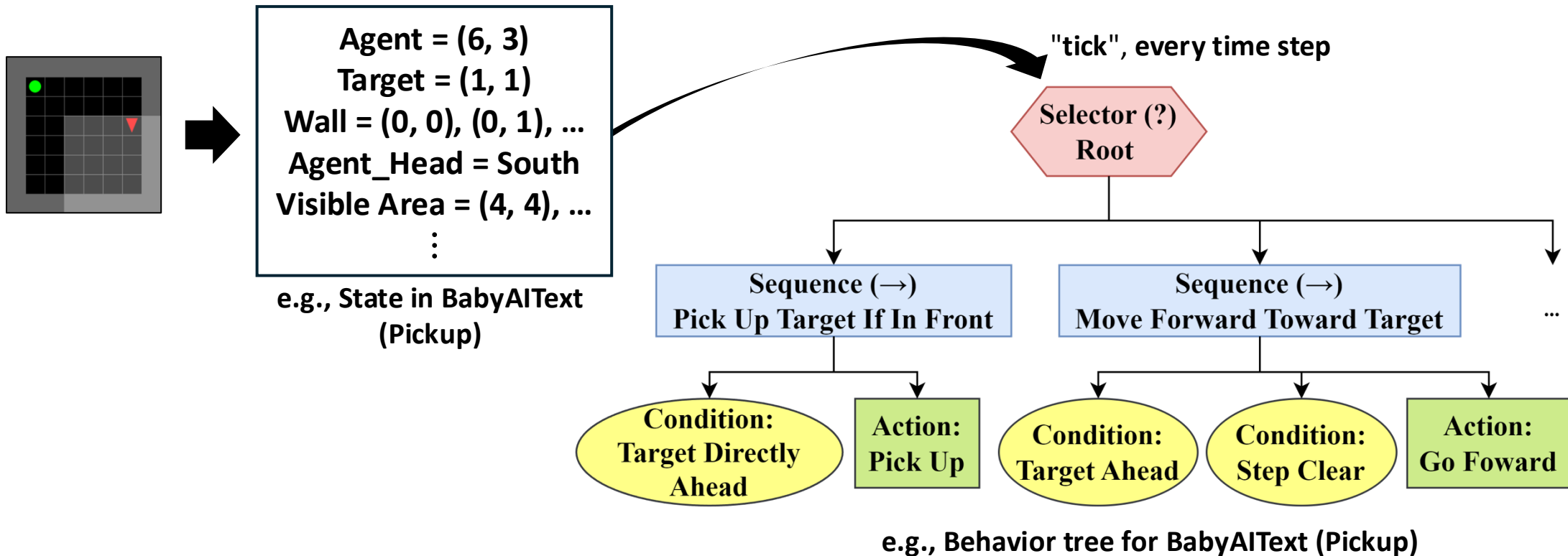
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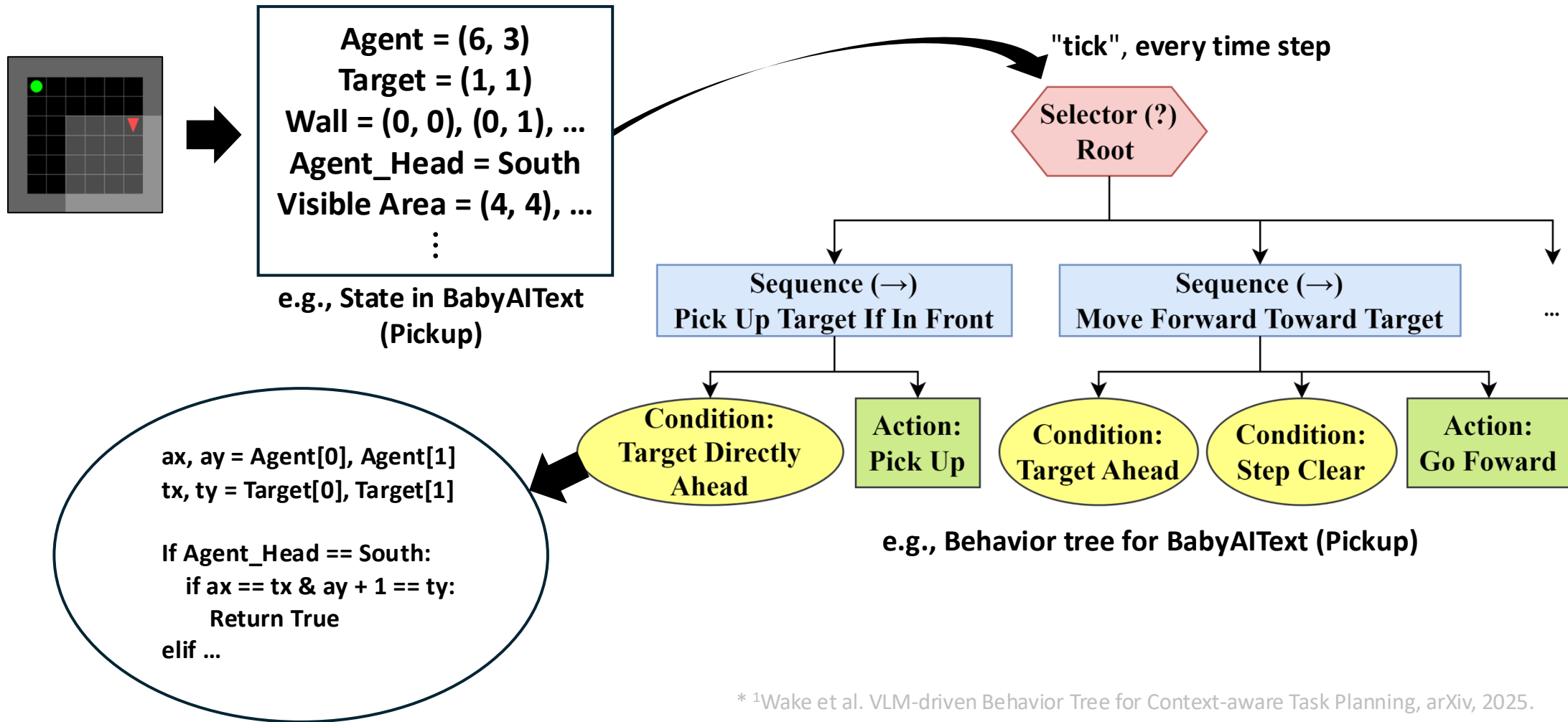
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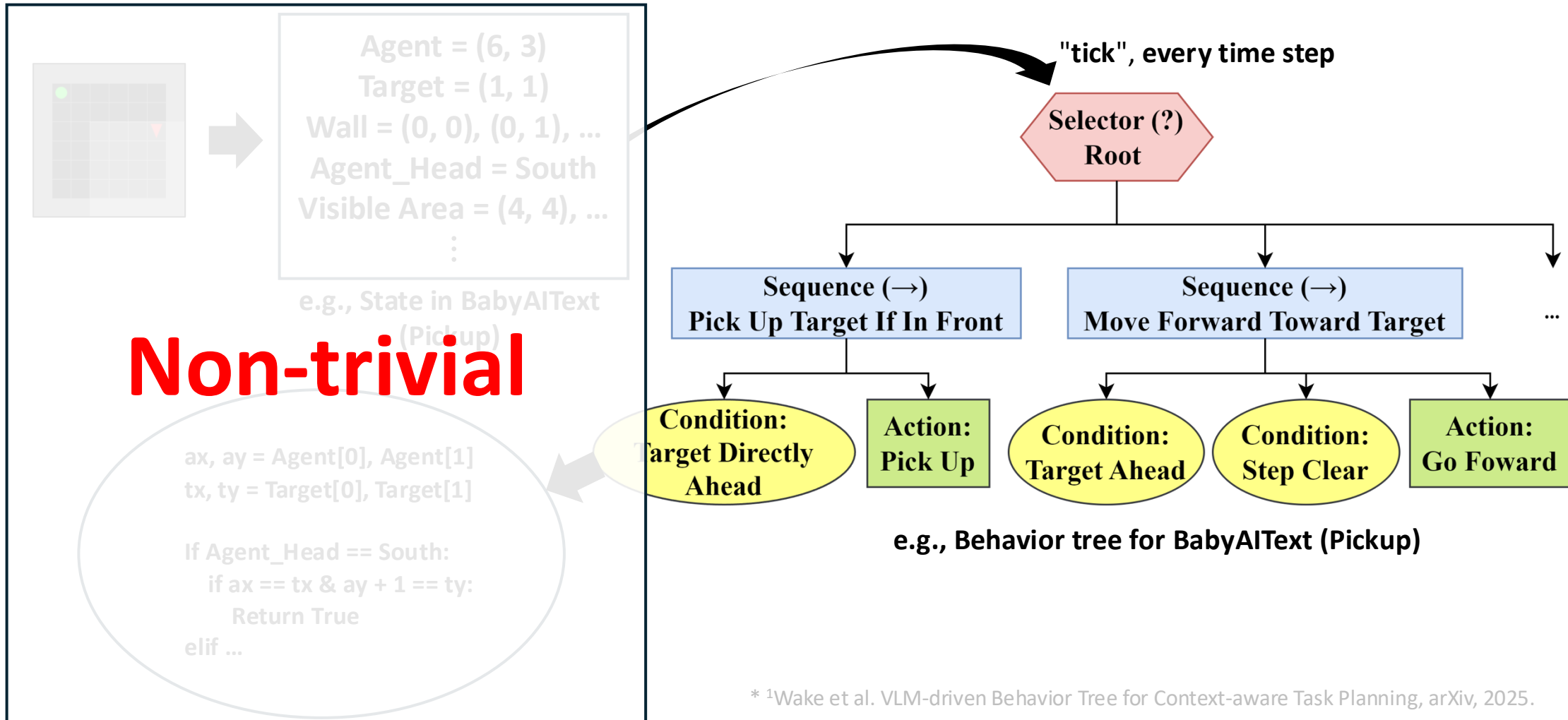
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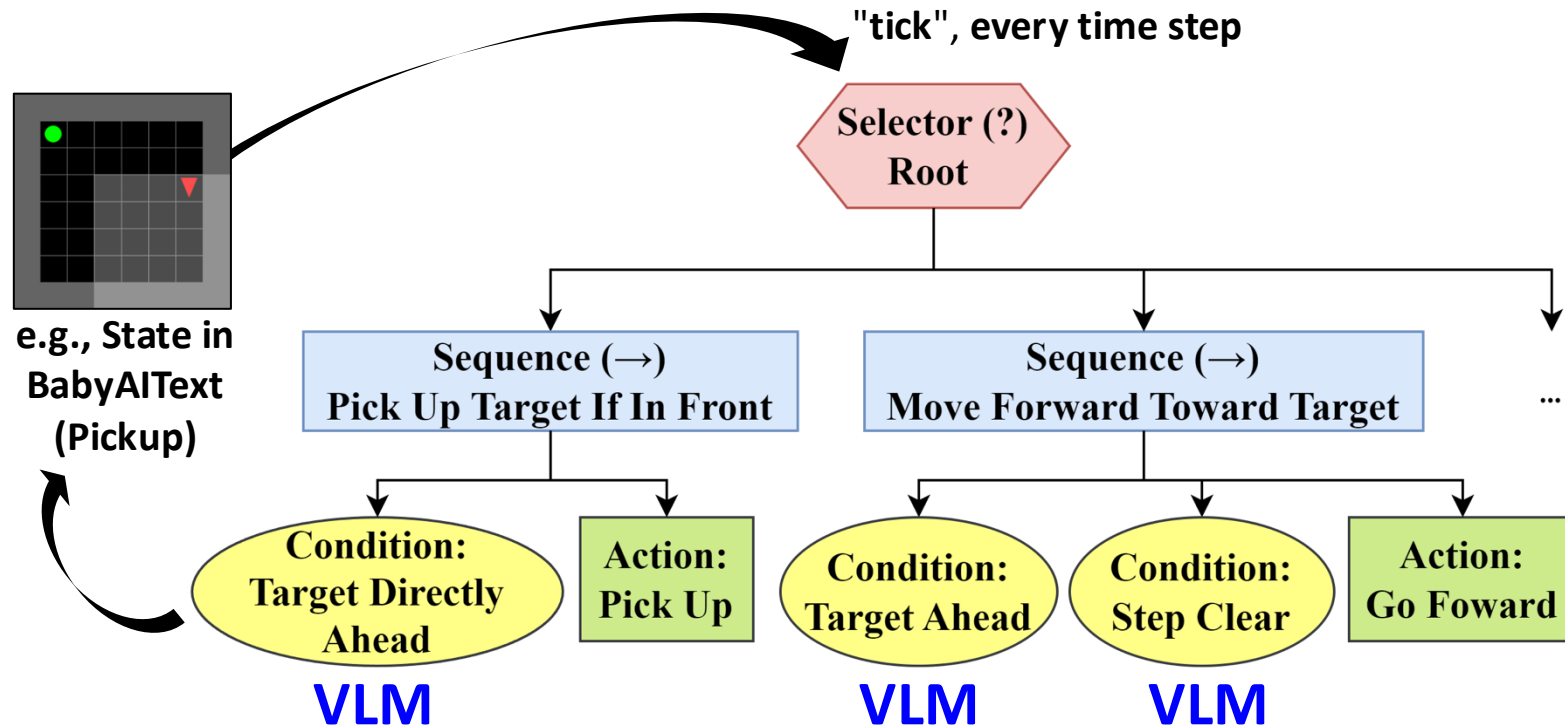
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- **Vision-Language Models (VLMs)** enable to process perceptual inputs (e.g., images or language)¹

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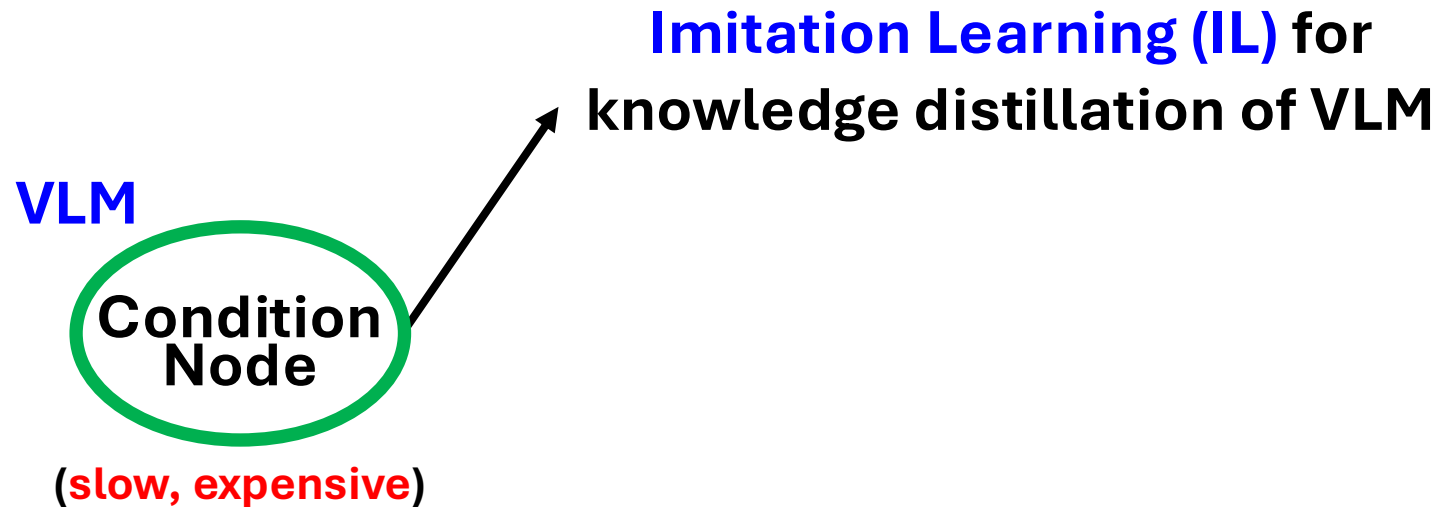
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- **Vision-Language Models (VLMs)** enable to process perceptual inputs (e.g., images or language)¹ but they are **slow** and **expensive** for real-time use.
(e.g., 1-hour control with 20 ms tick intervals = 180,000 API calls = Cost↑)

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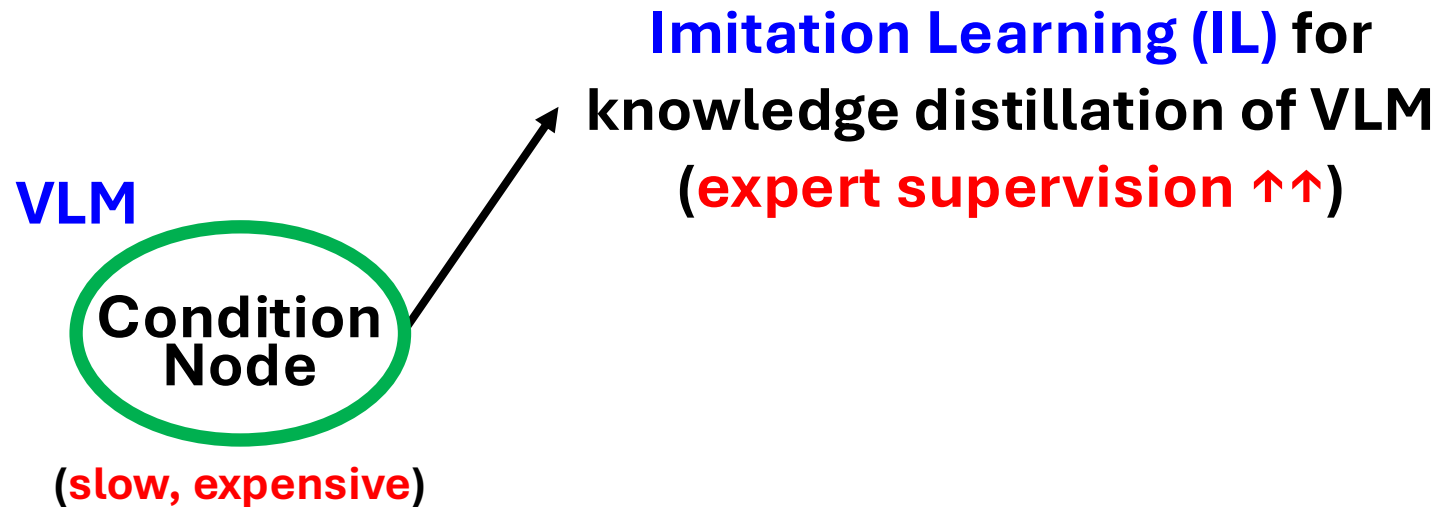
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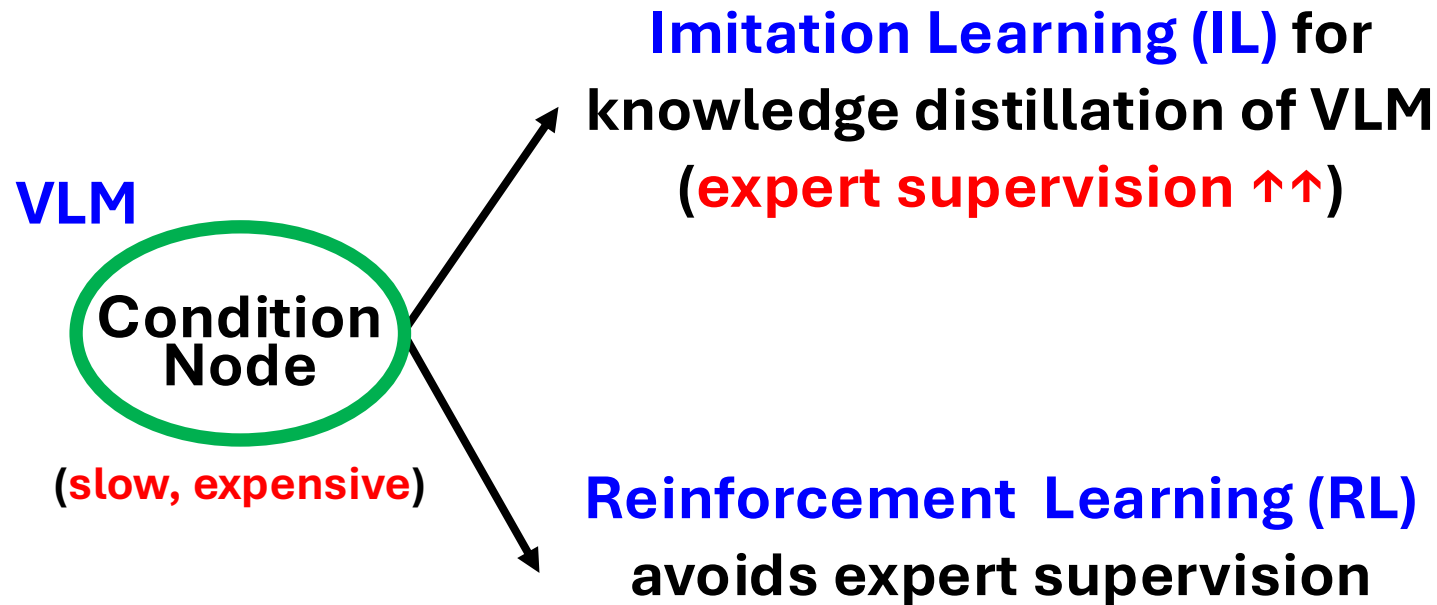
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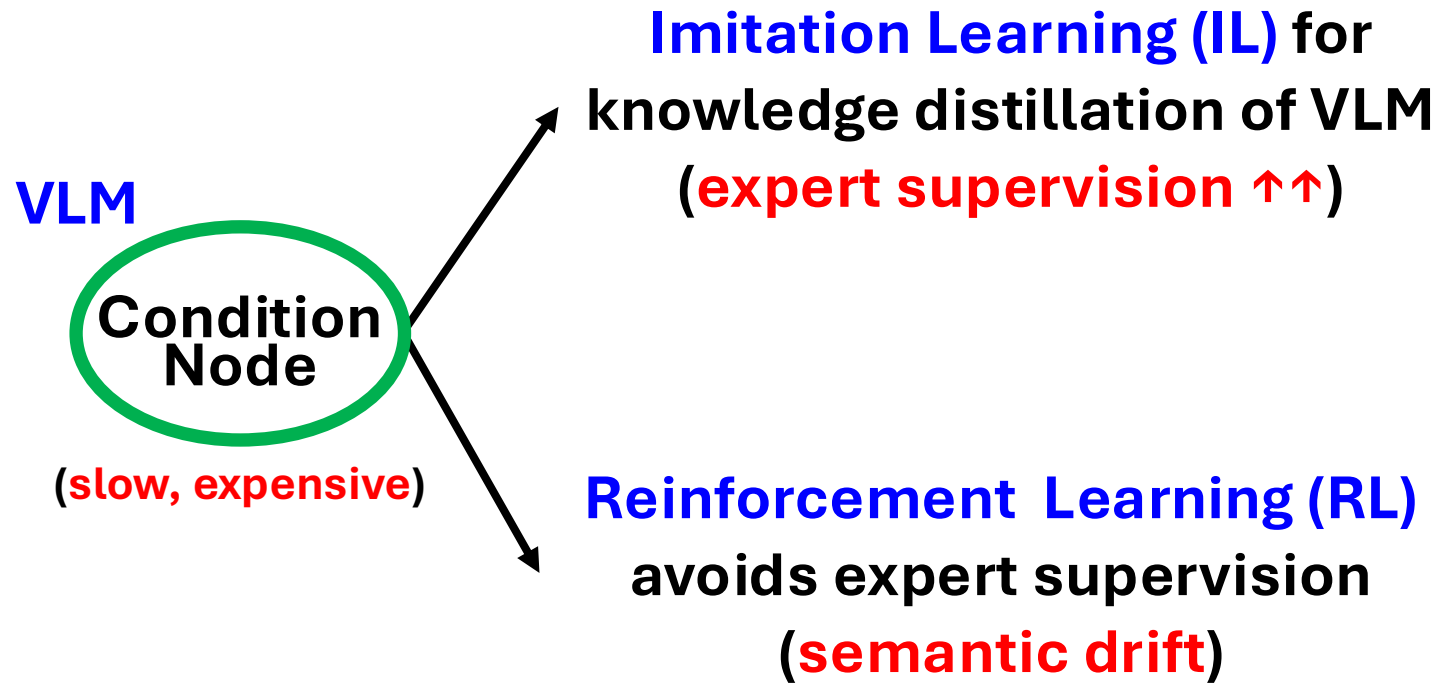
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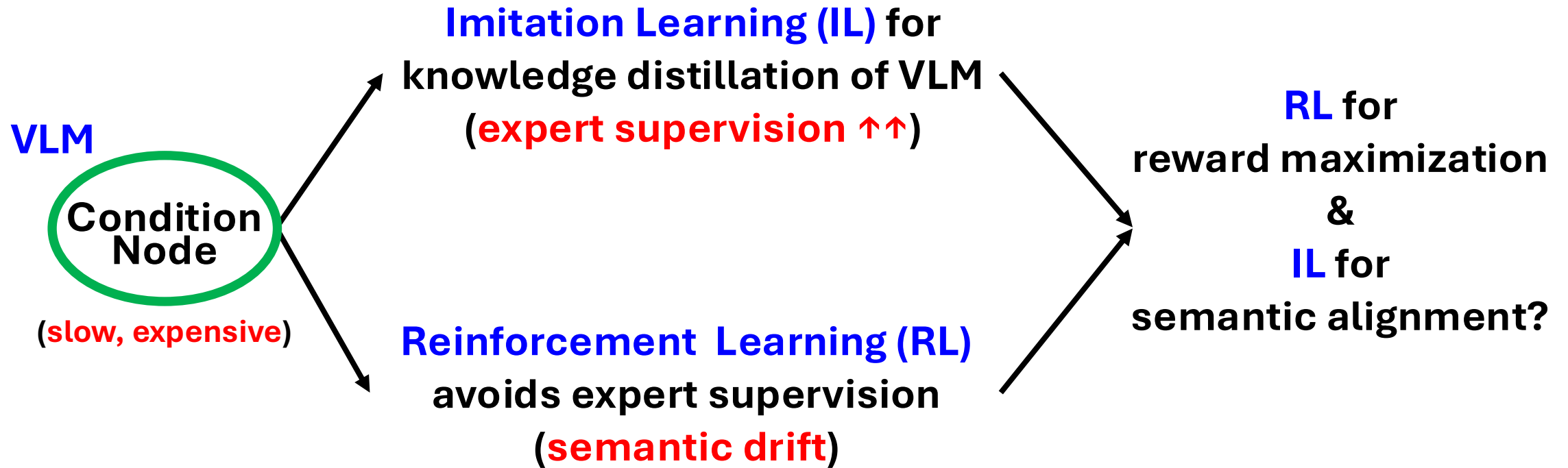
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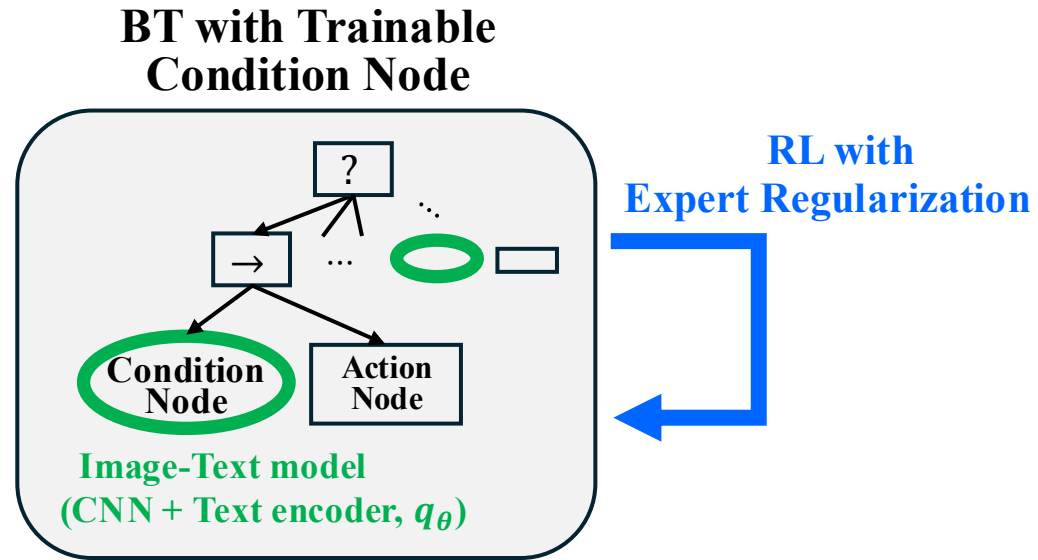
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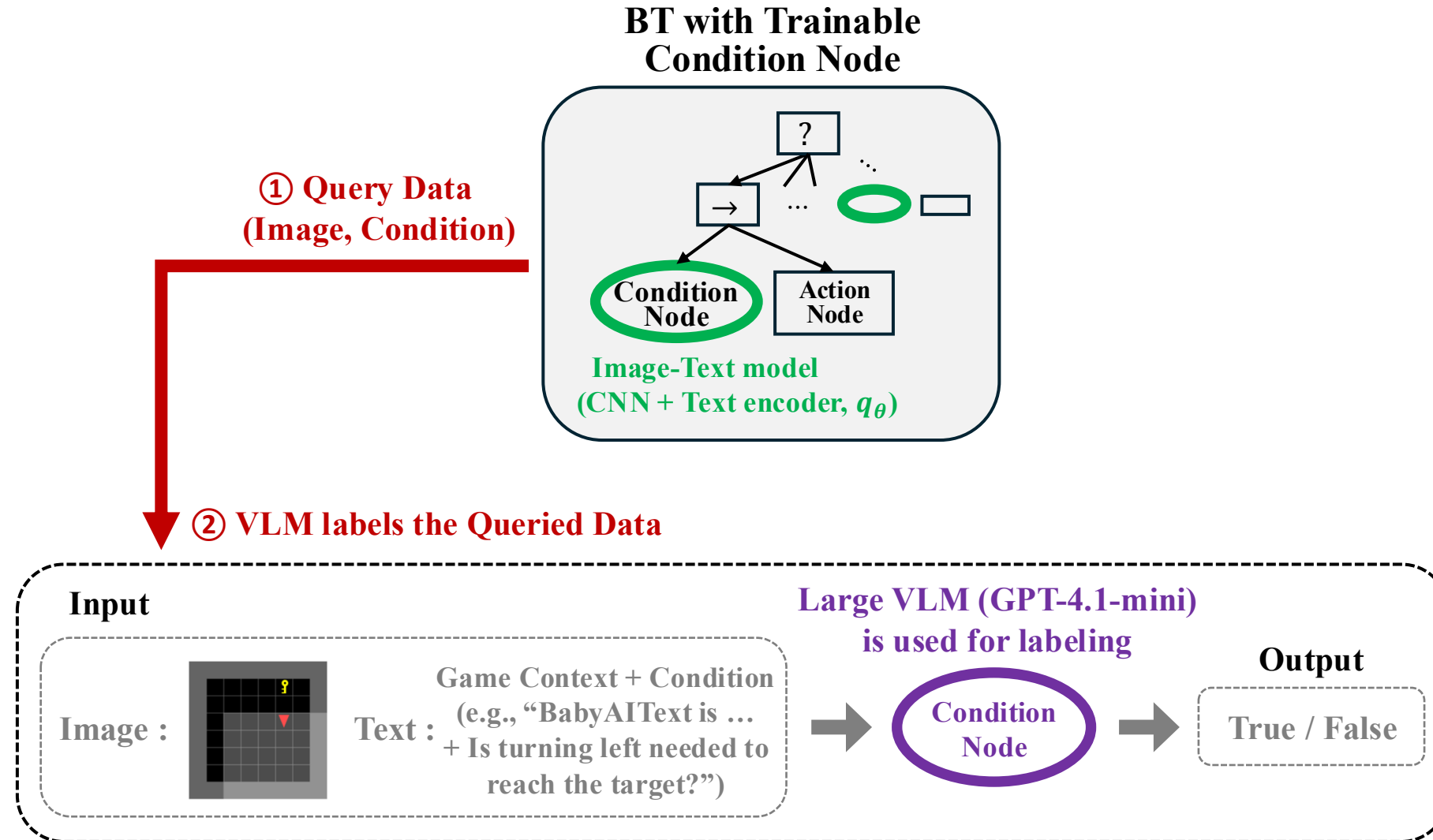


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■ Condition-node Expert-regularized Reinforcement Learning (CERL)

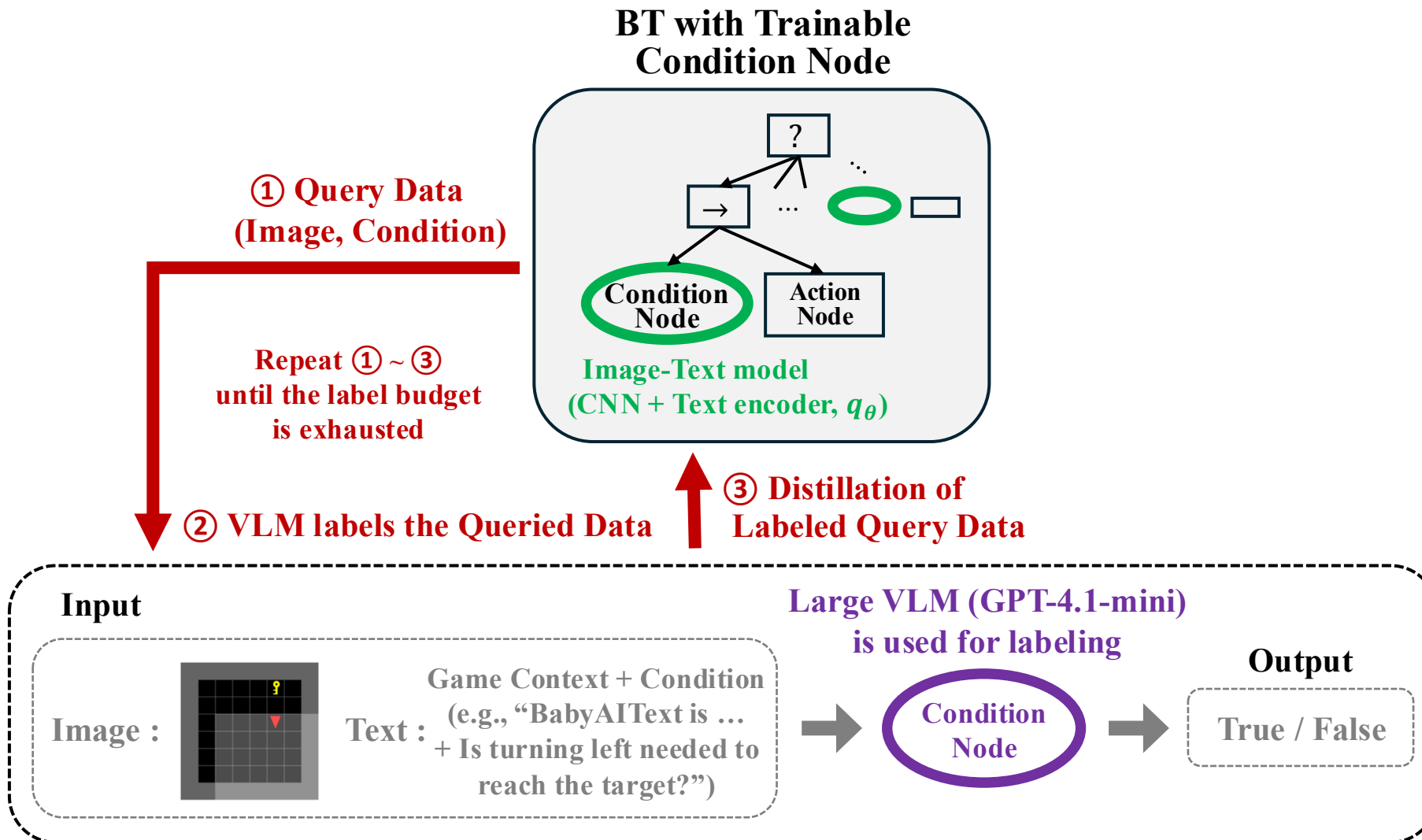


Condition-node Expert-regularized Reinforcement Learning (CERL)



Step 1. Expert Labeling
(①, ②): VLM generates labels for language queries on images.

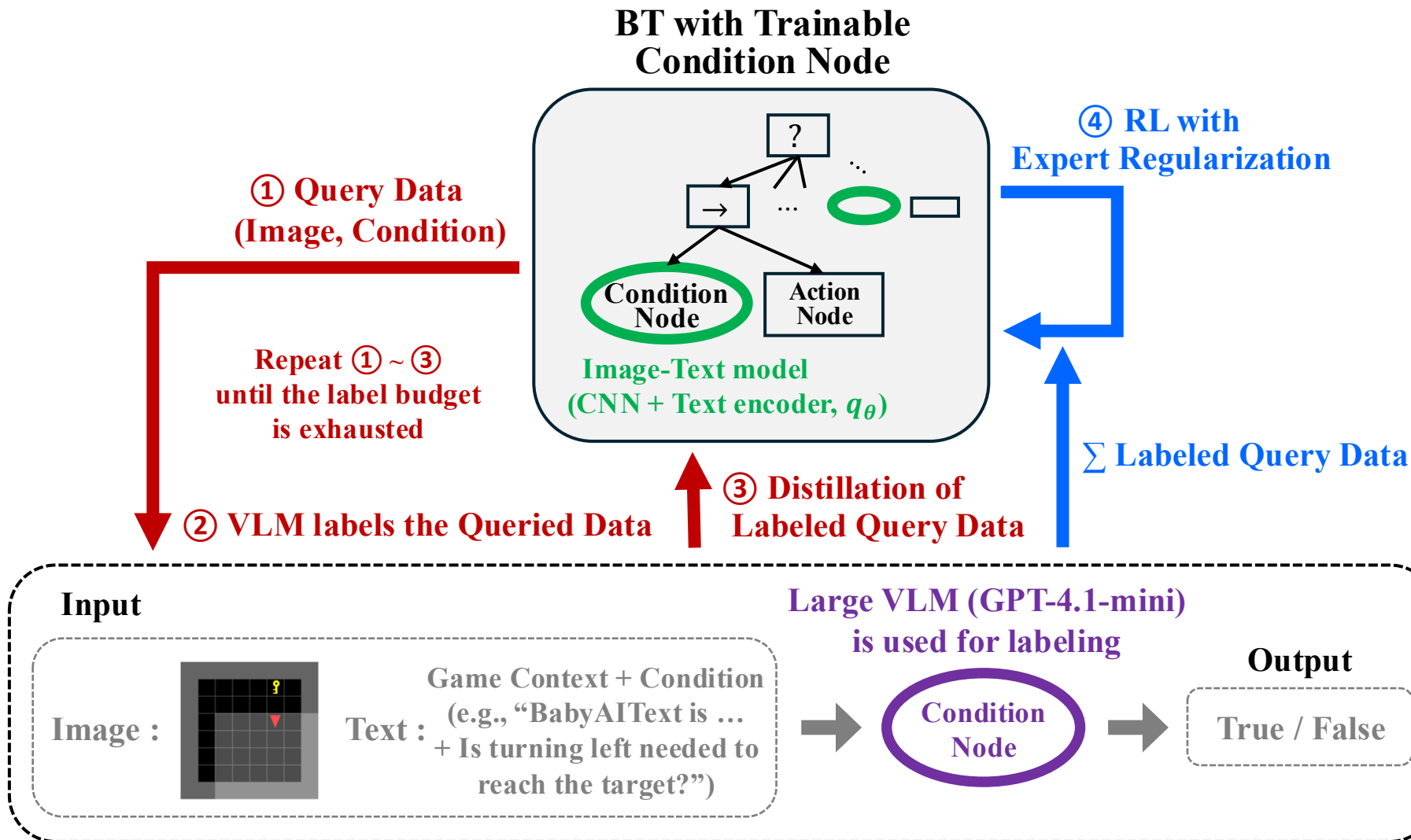
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Step 1. Expert Labeling (①, ②): VLM generates labels for language queries on images.

Step 2. Warm Start (③): The condition node is initialized via Behavior Cloning or DAgger.

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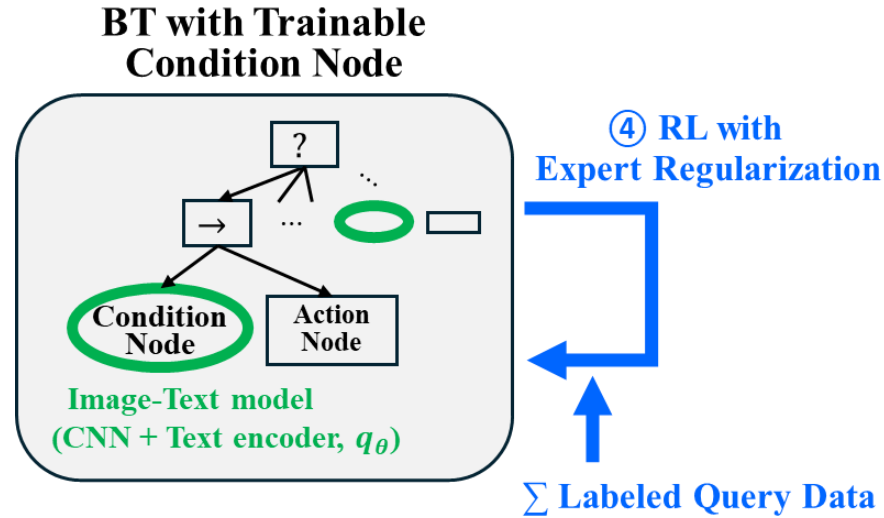


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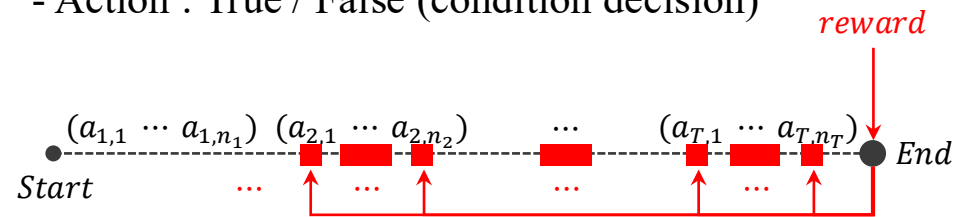
Step 3. Expert-Regularized RL (④): PPO optimizes rewards while staying aligned with expert labels used for initialization.

Condition-node Expert-regularized Reinforcement Learning (CERL)



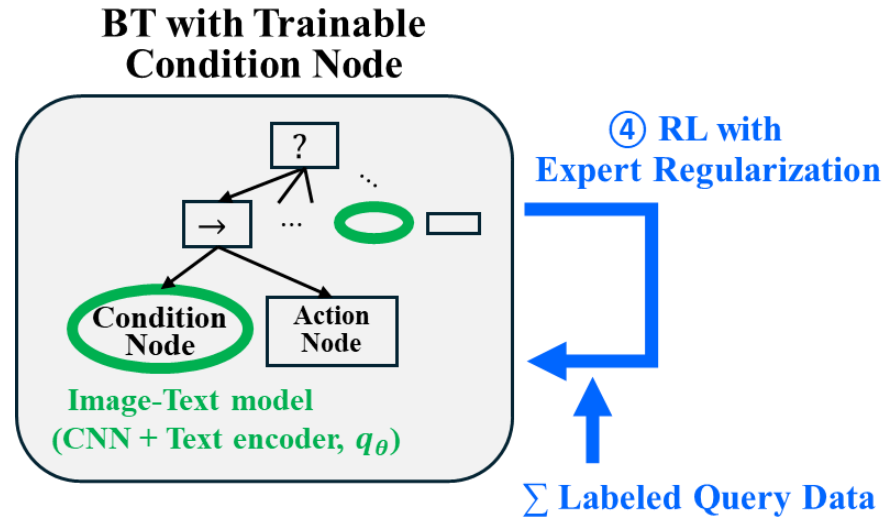
1) Naive RL (RL_b)

- State : Image + Text
- Action : True / False (condition decision)



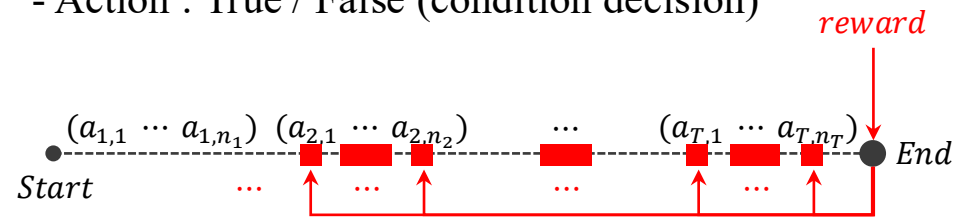
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- ⇒ Credit assignment issues

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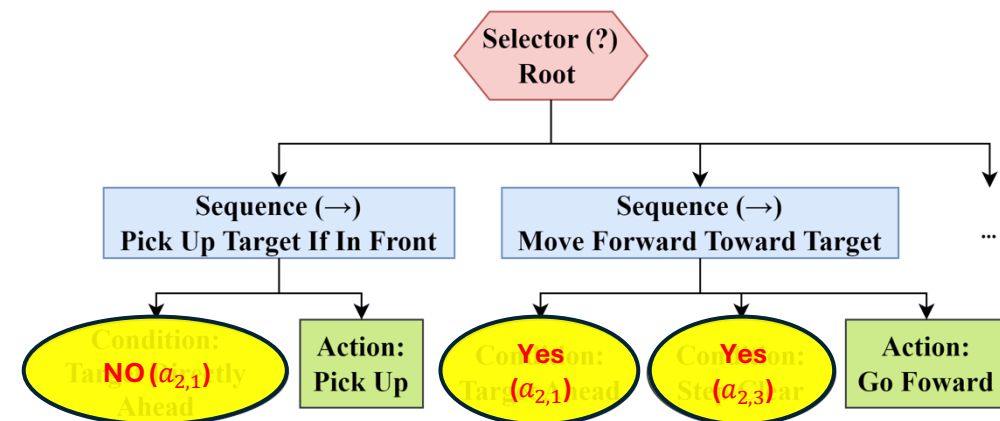


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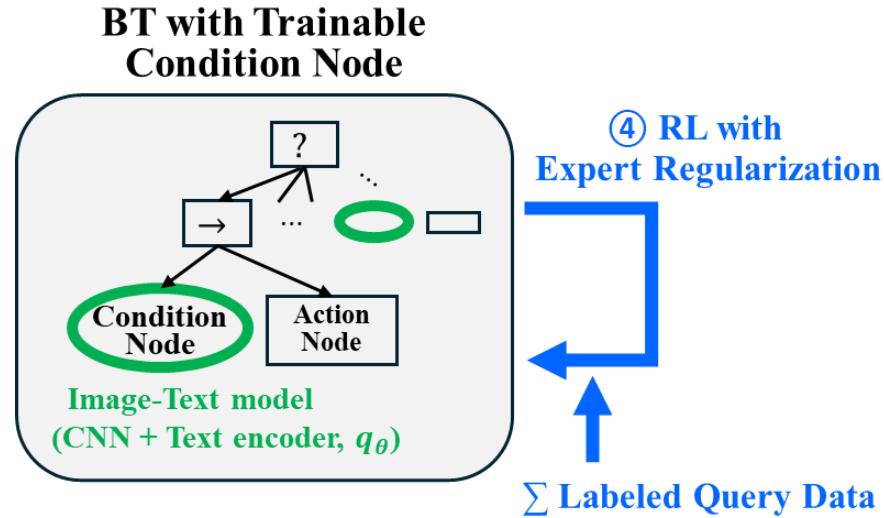
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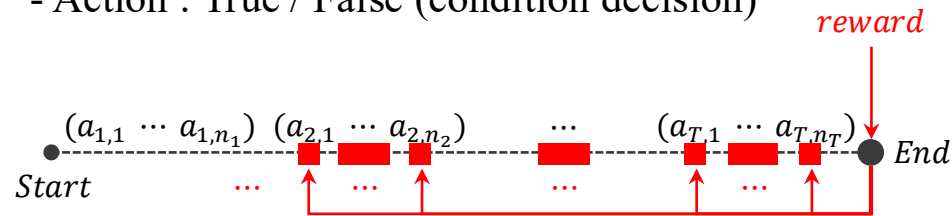


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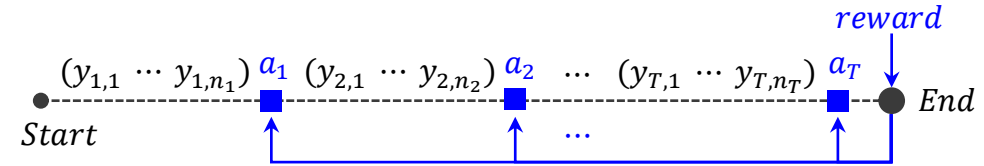
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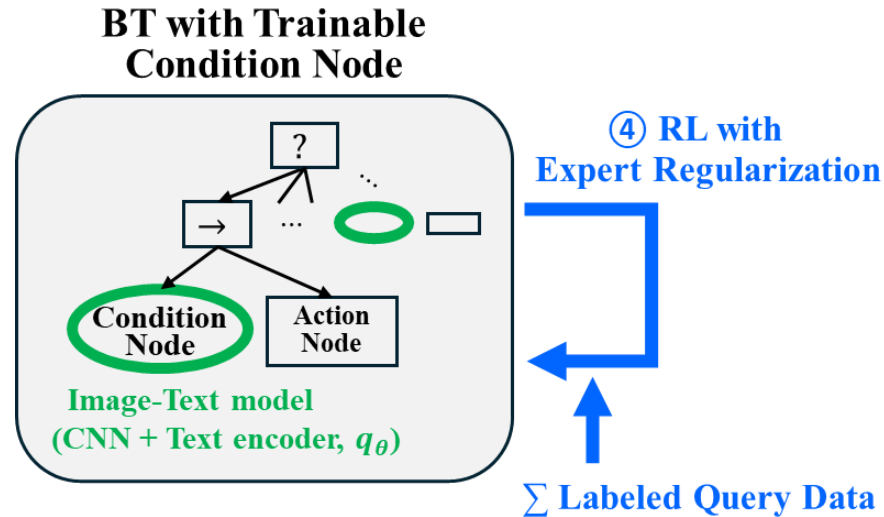
2) Factorized RL (RL_f (ours))

- State : Image
- Action : **Primitive environmental action** (e.g., Left, Right, etc)



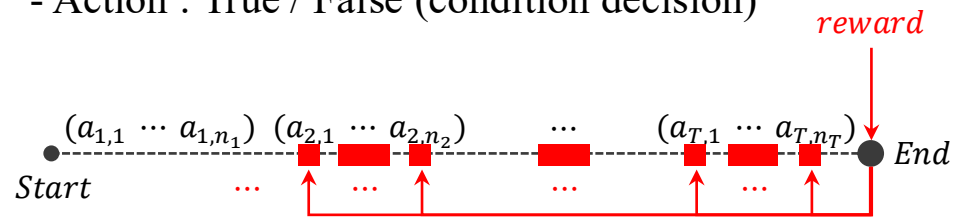
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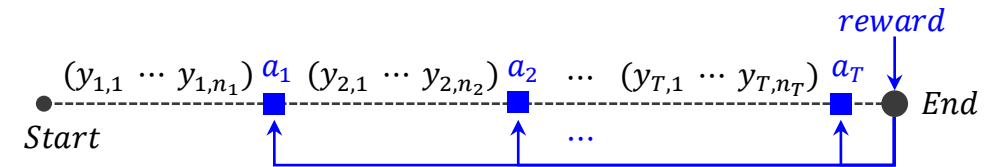
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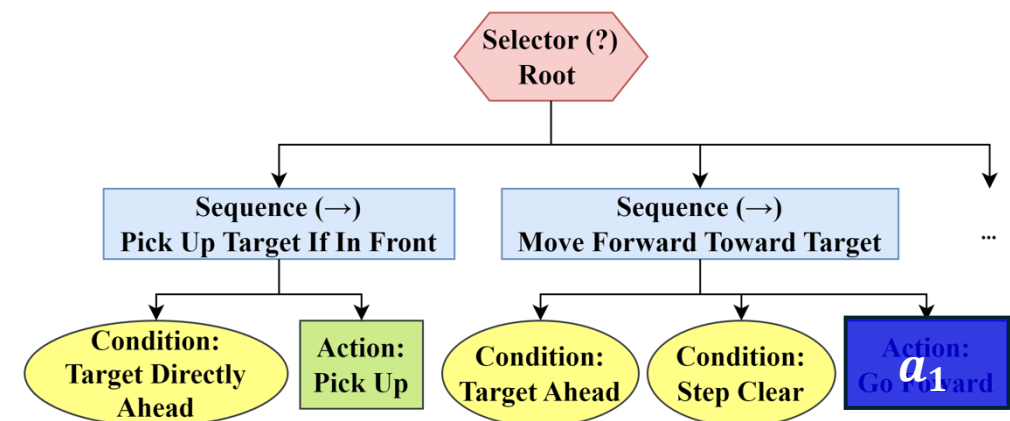
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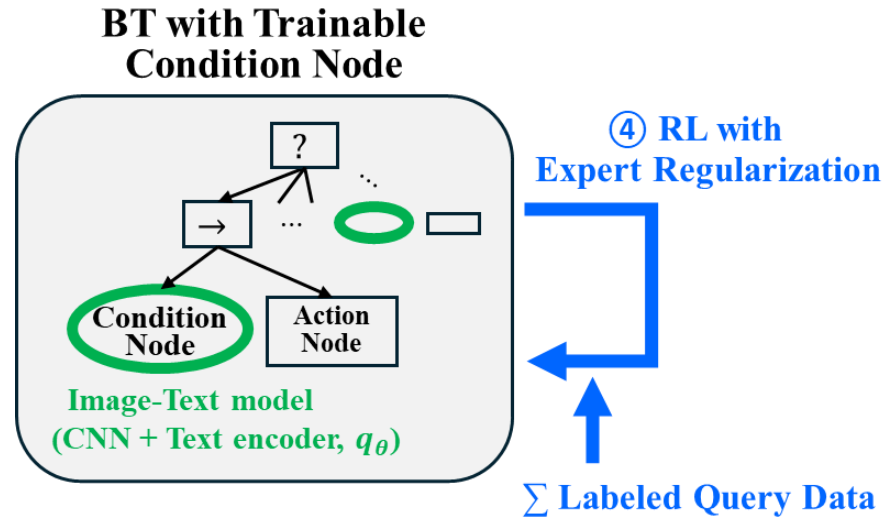
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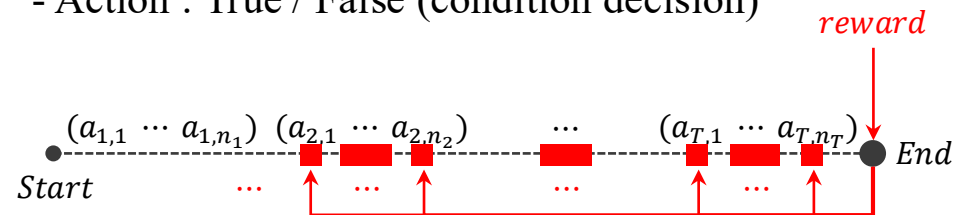


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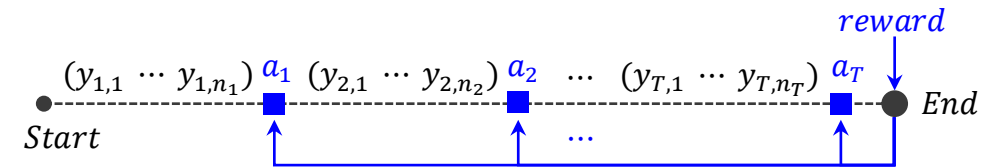
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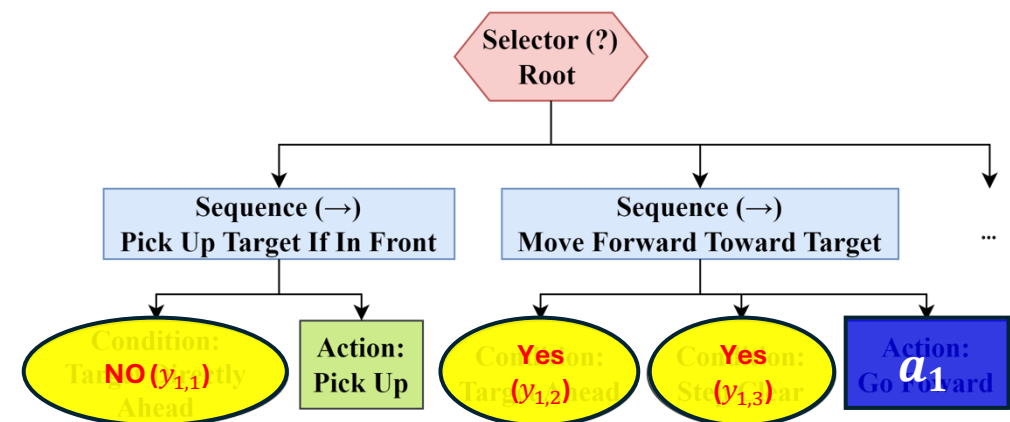
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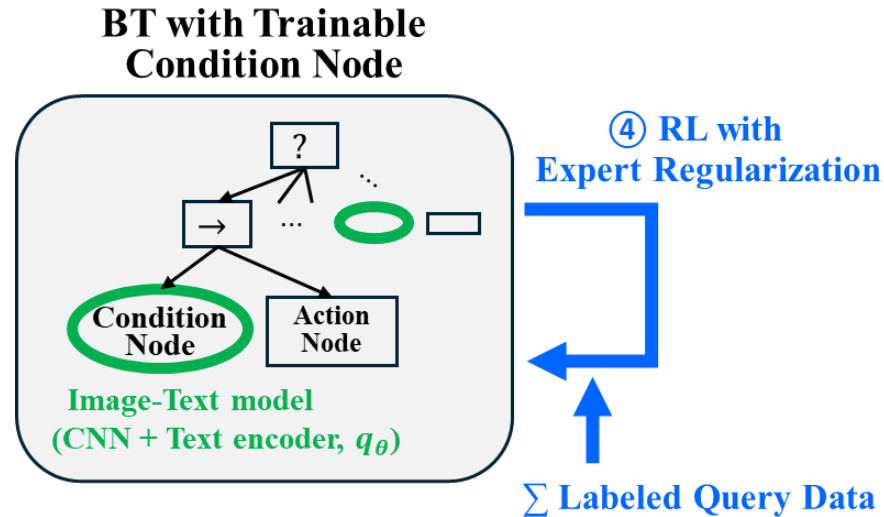
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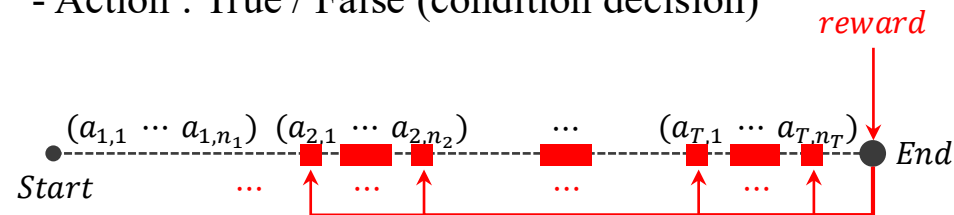


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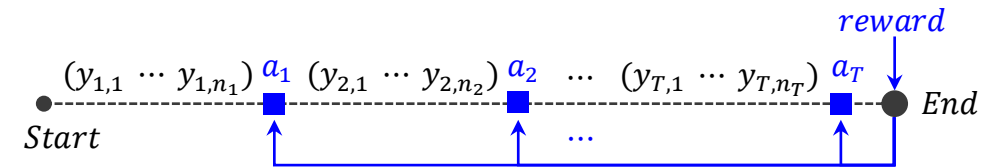
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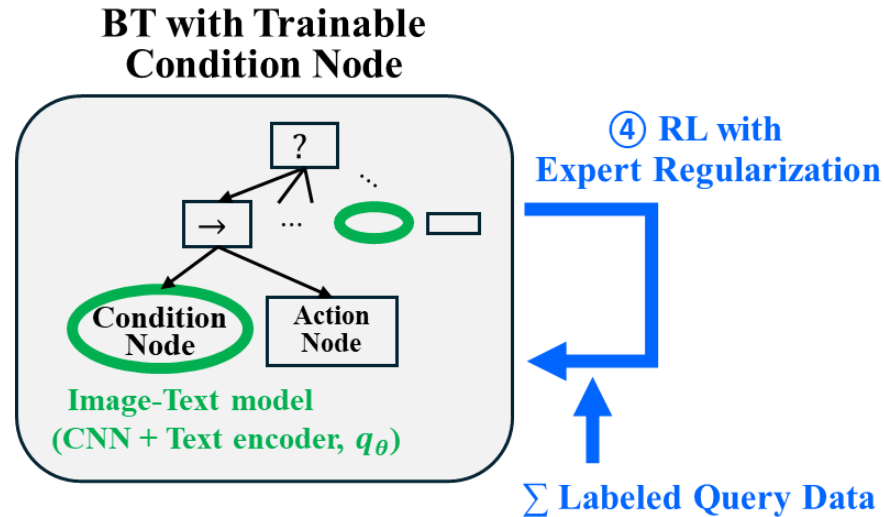
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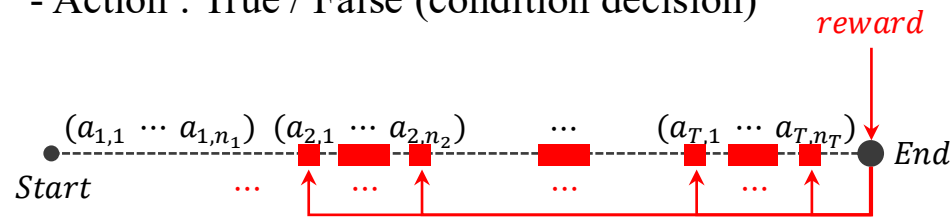
$$\pi(a_t | s_t) = \prod_{i=1}^{n_t} q_\theta(y_{t,i} | s_t, q_i)$$

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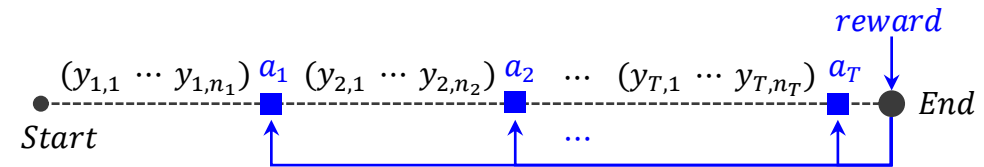
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$$\pi(a_t | s_t) = \prod_{i=1}^{n_t} q_\theta(y_{t,i} | s_t, q_i)$$

Theorem 1. Policy gradient for BT-driven policies

$$\Rightarrow \nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T \sum_{i=1}^{n_t} A_t \nabla_\theta \log q_\theta(y_{t,i} | s_t, q_i) \right]$$

⇒ Optimized with Expert Regularization
(Cross Entropy between $q_\theta(\cdot)$ and expert labels)

■ Experiment settings

1) Environment (Spare reward settings)

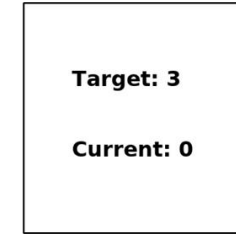
- NumberLine, EZPoints, BlackJack from GymCards
- 4x4, 8x8 from FrozenLake
- Goto, Pickup from BabyAIText

2) Models

- IL baselines : Behavior Cloning (BC) and Dagger (DA)
- RL baselines : Baseline RL (RL_b), Factorized RL (RL_f (ours))
- IL + RL (ours) : RL_f with expert regularization (weights: 1.0 and 0.1)
- Ablations : Variants removing initialization or regularization, and DAgger

3) Evaluation schemes (Averaged over four random seeds)

- Episodic Success Rate : proportion of successful episodes out of 100 episodes
- Decision Accuracy : agreement between learned and expert (VLM) condition decisions
- Inference Time : mean runtime per episode
- Model Size : number of parameters of the models



NumberLine



BlackJack



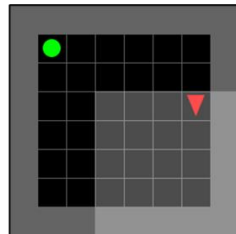
EZPoints



FrozenLake
4x4



FrozenLake
8x8



BabyAIText
Goto/Pickup

Experiment results

Metrics	Model		GymCards			FrozenLake		BabyAIText		Avg
			NL	BJ	EZP	4x4	8x8	Goto	Pickup	
Success Rate (%) ↑	Expert (GPT-4.1-mini)		100.00	36.75	78.00	98.25	82.00	99.75	99.50	84.89
	IL	BC	100.00	40.25	84.50	65.50	13.75	78.75	35.50	59.75
		DA	100.00	36.25	44.50	69.75	19.75	66.25	15.75	50.32
	RL	RL _b	100.00	28.00	0.00	0.00	0.00	5.25	0.75	19.14
		RL _f (ours)	100.00	44.00	0.00	97.75	0.00	86.75	2.50	47.29
	IL + RL (ours)	init _{BC} + RL _f w/ ER _{1.0}	100.00	42.50	89.00	95.50	58.50	95.25	65.00	77.96
		init _{BC} + RL _f w/ ER _{0.1}	100.00	44.25	85.00	97.25	66.50	95.50	70.75	79.89
	Ablation	RL _f w/ ER _{0.1}	100.00	40.25	0.00	99.00	21.75	98.75	9.50	52.75
		init _{BC} + RL _f	100.00	41.75	78.00	95.00	59.25	95.50	71.00	77.21
		init _{DA} + RL _f w/ ER _{0.1}	100.00	45.25	79.00	96.50	69.25	93.00	35.25	74.04
Accuracy (%) ↑	Expert (GPT-4.1-mini)		100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	IL	BC	100.00	91.50	98.90	91.54	81.88	83.98	84.67	90.35
		DA	100.00	93.56	91.85	91.47	87.09	83.03	79.74	89.53
	RL	RL _b	100.00	57.84	53.57	77.78	29.13	69.62	53.75	63.10
		RL _f (ours)	95.71	58.52	61.19	73.09	34.66	76.34	73.05	67.51
	IL + RL (ours)	init _{BC} + RL _f w/ ER _{1.0}	100.00	93.08	99.21	93.45	82.89	87.07	87.53	91.89
		init _{BC} + RL _f w/ ER _{0.1}	100.00	85.68	98.62	88.59	78.37	87.33	87.94	89.50
	Ablation	RL _f w/ ER _{0.1}	100.00	75.95	86.00	87.78	76.09	79.77	78.51	83.44
		init _{BC} + RL _f	100.00	72.87	95.34	77.44	68.23	84.00	85.49	83.34
		init _{DA} + RL _f w/ ER _{0.1}	100.00	88.90	97.04	88.48	86.97	87.15	83.00	90.22

- IL+RLs (ours) infer **faster** at **~0.09 seconds** per episode, compared to over **63 seconds** for the expert and are much **smaller** with **6.6M** parameters versus the expert's **7B¹**.

Experiment results

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			NL	BJ	EZP	4x4	8x8	Goto	Pickup	
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		init _{BC} + RL _f w/ ER _{0.1}	100.00	85.68	98.62	88.59	78.37	87.33	87.94	89.50
	Ablation	RL _f w/ ER _{0.1}	100.00	75.95	86.00	87.78	76.09	79.77	78.51	83.44
		init _{BC} + RL _f	100.00	72.87	95.34	77.44	68.23	84.00	85.49	83.34
		init _{DA} + RL _f w/ ER _{0.1}	100.00	88.90	97.04	88.48	86.97	87.15	83.00	90.22

- IL+RLs (ours) infer **faster** at **~0.09 seconds** per episode, compared to over **63 seconds** for the expert and are much **smaller** with **6.6M** parameters versus the expert's **7B¹**.

Experiment results

Metrics	Model		GymCards			FrozenLake		BabyAIText		Avg
			NL	BJ	EZP	4x4	8x8	Goto	Pickup	
Success Rate (%) ↑	Expert (GPT-4.1-mini)		100.00	36.75	78.00	98.25	82.00	99.75	99.50	84.89
	IL	BC	100.00	40.25	84.50	65.50	13.75	78.75	35.50	59.75
		DA	100.00	36.25	44.50	69.75	19.75	66.25	15.75	50.32
	RL	RL _b	100.00	28.00	0.00	0.00	0.00	5.25	0.75	19.14
		RL _f (ours)	100.00	44.00	0.00	97.75	0.00	86.75	2.50	47.29
	IL + RL (ours)	init _{BC} + RL _f w/ ER _{1.0}	100.00	42.50	89.00	95.50	58.50	95.25	65.00	77.96
		init _{BC} + RL _f w/ ER _{0.1}	100.00	44.25	85.00	97.25	66.50	95.50	70.75	79.89
	Ablation	RL _f w/ ER _{0.1}	100.00	40.25	0.00	99.00	21.75	98.75	9.50	52.75
		init _{BC} + RL _f	100.00	41.75	78.00	95.00	59.25	95.50	71.00	77.21
		init _{DA} + RL _f w/ ER _{0.1}	100.00	45.25	79.00	96.50	69.25	93.00	35.25	74.04
Accuracy (%) ↑	Expert (GPT-4.1-mini)		100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	IL	BC	100.00	91.50	98.90	91.54	81.88	83.98	84.67	90.35
		DA	100.00	93.56	91.85	91.47	87.09	83.03	79.74	89.53
	RL	RL _b	100.00	57.84	53.57	77.78	29.13	69.62	53.75	63.10
		RL _f (ours)	95.71	58.52	61.19	73.09	34.66	76.34	73.05	67.51
	IL + RL (ours)	init _{BC} + RL _f w/ ER _{1.0}	100.00	93.08	99.21	93.45	82.89	87.07	87.53	91.89
		init _{BC} + RL _f w/ ER _{0.1}	100.00	85.68	98.62	88.59	78.37	87.33	87.94	89.50
	Ablation	RL _f w/ ER _{0.1}	100.00	75.95	86.00	87.78	76.09	79.77	78.51	83.44
		init _{BC} + RL _f	100.00	72.87	95.34	77.44	68.23	84.00	85.49	83.34
		init _{DA} + RL _f w/ ER _{0.1}	100.00	88.90	97.04	88.48	86.97	87.15	83.00	90.22

- IL+RLs (ours) infer **faster** at **~0.09 seconds** per episode, compared to over **63 seconds** for the expert and are much **smaller** with **6.6M** parameters versus the expert's **7B¹**.

■ Conclusion

1) Summary

- **Interpretable visual decision-making framework** via expert-regularized RL in Behavior Trees
- **Motivation:** Address the limitations of VLMs for interpretable sequential decision-making in BTs
- **Contribution:** A unified framework **integrating IL and RL** via expert regularization and **sampling-based policy gradients** for BT-driven policies from perceptual inputs

2) Future works

- **Automated BT synthesis** and adaptation
- Handling on **noisier expert supervision**



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Q&A

Thank you